

GENERAL SPECIFICATION G5.10

Basic Electrical Materials and Methods

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G5.10 BASIC ELECTRICAL MATERIALS AND METHODS

G5.10.1 General

(a) Introduction

- (i) General Specifications G5 Electrical Works, Particular Specification P5 Electrical Works, and Drawings for Electrical Works covers the work necessary for the complete electrical system that is fixed in position, connected, tested, safe and in fully functional working order. Supply, deliver, install, test and commission materials and equipment in accordance with these Specifications, Drawings, and all other documents with respect to this Contract.
- (ii) The work shall include provision of all labour and all materials necessary to form a complete installation and such tests, adjustments and commissioning as are prescribed in subsequent sections and as may otherwise be required to give an effective working installation to the satisfaction of the S.O..
- (iii) Supplying and installing all incidental items not actually shown or specified, but which are required by good practice, safe and reliable operation and maintenance to provide complete functional systems.
- (iv) The Contractor shall ensure that all requirements are strictly adhered to when carrying out work required by these Contract Documents, unless otherwise instructed by the S.O..

(b) References

- (i) Perform all work; supply, deliver and install all materials and equipment, and installation methods in accordance with, but not limited to, the latest applicable rules, regulations, requirements, local and international codes as listed in General Specification G5.2-Standards, Codes and Regulations.
- (ii) Conflicts, if any, that may exist between the above reference standards, codes and regulations will be resolved at the discretion of the S.O..

(c) Submittals- The Contractor shall liaise and consult with manufacturers and advise S.O on the critical information for O&M and if required, the manufacturers shall present the details to S.O..

- (i) Shop Drawings, Samples and Technical Data: As a minimum, submit the following items:
 - 1) Device boxes for use in hazardous areas;
 - 2) Junction and pull boxes used at or below grade;
 - 3) Large junction and pull boxes;
 - 4) Terminal junction boxes;
 - 5) Panel boards and circuit breaker data;
 - 6) Wiring devices;
 - 7) Control devices;
 - 8) Control relays;

- 9) Timers;
 - 10) Local Control Stations:
 - Panel layout; and
 - Detailed wiring diagrams.
 - 11) Fuses;
 - 12) Magnetic contactors;
 - 13) Intrinsic safety barriers;
 - 14) Fire stopping;
 - 15) Enclosures: Include enclosure data for products having enclosures; and
 - 16) All equipment shown on Drawings.
- (d) Equipment for Hazardous Areas
- (i) Materials and devices shall be specifically approved for hazardous areas of the zone shown and of a construction that will ensure safe performance when properly used and maintained.
- (e) Spares
- (i) Provide spare parts, materials and special tools as listed in the Schedule of Spares, Special Tools and Consumable Items as specified in General Specification G2.28- Spare Parts, Tools and Test Equipment.

G5.10.2 Products

- (a) Metering Facilities (Or Metering Kiosks)
- (i) Supply, install, test and commission materials as required by SP Group for installation of Metering Kiosks, billing metering equipment, service conductors and mounting of utility company equipment.
 - (ii) Comply with all requirements of SP Group and EMA.
- (b) Outlet and Device Boxes
- (i) Sheet Steel: One-piece drawn type, electro-galvanised finish complying with BS EN 10277, BS 1449 and BS EN 10095.
 - (ii) Cast Metal:
 - 1) Box: Malleable iron of circular pattern with spout;
 - 2) Cover: Gasketed, weatherproof, malleable iron, with stainless steel screws;
 - 3) Hubs: Threaded;
 - 4) Lugs: Cast mounting; and
 - 5) Finish: Electro-galvanised finish complying with BS EN 10277, BS 1449, and BS EN 10095.

- (iii) Cast Aluminium:
 - 1) Material:
 - Box: Cast, copper-free aluminium; and
 - Cover: Gasketed, weatherproof, cast copper-free aluminium with stainless steel screws.
 - 2) Hubs: Threaded; and
 - 3) Lugs: Cast mounting.
- (iv) PVC-Coated Cast Metal:
 - 1) Type: One-piece;
 - 2) Material: Malleable iron, with electro-galvanised finish complying with BS EN 10277, BS 1449, and BS EN 10095; and
 - 3) Coating:
 - All exterior surfaces, 1 mm PVC; and
 - All interior surfaces, 0.05 mm urethane.
- (v) NON-METALLIC:
 - 1) Box: PVC of an approved type complying with BS 4607'
 - 2) Circular construction with mounting lugs external to the box;
 - 3) Cover: PVC, weatherproof, with stainless steel screws; and
 - 4) Colour: White to match conduit unless approved by S.O..
- (c) Junction Boxes and Pull Boxes
 - (i) Outlet Boxes Used as Junction or Pull Box: As specified under Article Outlet and Device Boxes.
 - (ii) Large Sheet Steel Box:
 - 1) Degree of Protection: IEC IP 55, minimum;
 - 2) Box: Code-gauge, electro-galvanised steel;
 - 3) Cover: Full access, screw type; and
 - 4) Machine Screws: Corrosion-resistant.
 - (iii) Large Cast Metal Box:
 - 1) Degree of Protection: IEC IP 65, minimum;
 - 2) Box: Cast malleable iron, electro-galvanised finished, with drilled and tapped conduit entrances and exterior mounting lugs;
 - 3) Cover: Non-hinged with screws;
 - 4) Gasket: Neoprene; and
 - 5) Hardware and Machine Screws: BS EN 10088, Type 316L stainless steel.

- (iv) Large Cast Metal Box, Hazardous Locations:
 - 1) Degree of Protection: IEC IP 66, minimum;
 - 2) Conform to requirements of IEC 60079 as required for hazardous location zone involved and suitable rated with EX de;
 - 3) Box: Cast ferrous metal, electro-galvanise finished, or copper-free aluminium with drilled and tapped conduit entrances;
 - 4) Cover: Non-hinged with screws; and
 - 5) Hardware and Machine Screws: BS EN 10088, Type 316L stainless steel.
- (v) Large Cast Aluminium Box:
 - 1) Degree of protection: IEC IP 65;
 - 2) Box: Cast copper-free aluminium, with drilled and tapped conduit entrances and exterior mounting lugs;
 - 3) Cover: Non-hinged;
 - 4) Gasket: Neoprene; and
 - 5) Hardware and Machine Screws: BS EN 10088, Type 316L stainless steel.
- (vi) Large Stainless Steel Box:
 - 1) Degree of Protection: IEC IP 65;
 - 2) Box: 1.6 mm (14-gauge) minimum, BS EN 10088, Type 316L stainless steel;
 - 3) Cover: Hinged with fast operating clamps;
 - 4) Hardware and Machine Screws: BS EN 10088, Type 316L stainless steel; and
 - 5) Mounting feet external to the enclosure.
- (vii) Large Non-Metallic Box:
 - 1) Degree of Protection: IEC IP 65;
 - 2) Box: High-impact, fibreglass-reinforced polyester or engineered thermoplastic, with stability to high heat;
 - 3) Cover: Hinged with clamps or screws;
 - 4) Hardware and Machine Screws: Type 316 stainless steel; and
 - 5) Conduit hubs and mounting lugs.
- (viii) Overground (OG) Box:
 - 1) The OG box shall be outdoor type;
 - 2) All components of OG box shall be accommodated in a weatherproof housing to IP 56 of robust construction;
 - 3) The OG box shall be provided with a watershed top;
 - 4) The housing shall be constructed of stainless steel grade 316 of thickness of not less than 2.5 mm and hairline finish;

- 5) Adequate ventilation shall be provided to permit natural circulation of air;
 - 6) The ventilation apertures shall be suitably screened to prevent entry of vermin and other foreign bodies;
 - 7) The OG box shall be provided with a roof of ample strength and suitable for mounting on a concrete/brick footing;
 - 8) Hinged door with lock shall be provided at the front of the OG box;
 - 9) The door locks and keys for all distribution boxes shall be similar and interchangeable; and
 - 10) Three sets of keys shall be provided for each OG box.
- (d) Telecommunication Facilities
- (i) Supply, deliver, install, test and commission materials and systems, cabling facilities, and space required to meet IMDA COPIF.
- (e) Wiring Devices
- (i) Switches:
 - 1) Type: Conform to IEC 60669;
 - 2) Rating: 20 amperes, 240 volts;
 - 3) Operation: Quick-make, quick-break;
 - 4) Single-pole or two-way as required;
 - 5) Moulded ivory with chrome plated brass cover plate; and
 - 6) Switches connected to an emergency supply shall be distinctively marked with a red dot permanently engraved on the cover.
 - (ii) Power Socket Outlet:

All power and switched socket outlets (industrial and domestic type, single phase or three phases shall be provided with protection covers and good quality padlock or number lock suitable for the applications. Contractor shall provide the matching plug for each power and socket outlets from the same manufacturer.

 - 1) 13-Ampere Outlet:
 - Three-pin, earthing type, conforming to SS 145;
 - 13-amperes switched socket outlet;
 - Rectangular-pin socket;
 - 13-amperes single-pole rocker type micro switch; and
 - Moulded ivory with chrome plated brass cover plate.
 - 2) 13-Ampere Outlet, Weatherproof:
 - Three-pin, earthing type, conforming to SS 145;
 - Rectangular-pin socket;

- 13-amperes single-pole rocker type micro switch; and
 - Metal-clad with degree of protection of IP 54.
- 3) 15-Ampere Socket Outlet:
- Three-pin, earthing type, conforming to SS 472;
 - Round-pin socket; and
 - Chrome plated brass cover plate.
- 4) 30-Ampere Three Phase Socket Outlet:
- 4- Pole 5 wire 400 V complete with equipment earthing;
 - Power interrupting locking device;
 - Shall be heavy duty moulded of rugged arc resistant;
 - Positive locking;
 - Self-closing and ensure protection from electrical hazard by closing off live connection when the cap is removed; and
 - Pad lockable.
- 5) 20- to 30-Amperes, 230-Volts Isolators:
- Conform to IEC 60669 and IEC 60947;
 - Rated 20-and 30-amperes, 230 volts, 50-Hz;
 - Integral earthing and power interrupting locking device;
 - Manufactured of heavy-duty, moulded, arc-resistant melamine; and
 - Self-closing cap.
- 6) 20- and 30-Amperes, 400-Volts Isolators:
- Conform to IEC 60669 and IEC 60947;
 - Rated 20- and 30-amperes, three-phase, four-wire, five-pole, 400 volts, 50-Hz;
 - Integral earthing and power interrupting locking device;
 - Manufactured of heavy-duty, moulded, arc-resistant melamine; and
 - Self-closing cap.
- 7) Power outlets connected to an emergency supply shall be distinctively marked with a red dot permanently engraved on the cover.
- 8) All power and switched socket outlets (Industrial or domestic type) shall be provided with protection covers and good quality padlocks or number locks suitable for the application. Contractor shall provide the matching plug, which is of the same manufacturer of outlets.
- (iii) Receptacle, Explosion Proof:
- 1) Dead front, interlocked, circuit breaking;
 - 2) Electrical Ratings: 20 to 63 amperes, up to 600 volts;

- 3) Hazardous Area Ratings: IEC 60079; and
 - 4) Provide matching plug with each receptacle.
- (f) Circuit Breaker, Individually Mounted
- (i) Moulded Case Circuit Breakers (MCCB):
 - 1) Moulded case, thermal-magnetic, quick make, and quick break, trip-free, of the indicating type showing OFF/ON and tripped position of the operating handle;
 - 2) In compliance with IEC 60947-1 and IEC 60947-2;
 - 3) Ultimate Breaking Capacity (Icu): 65 kA (minimum) at 415 V, AC;
 - 4) Ics not less than 100%% of Icu;
 - 5) Multi-pole circuit breakers designed to automatically open all poles when an overload occurs on one pole;
 - 6) Suitable for mounting in any position;
 - 7) Locking provisions for padlocking breaker in open position;
 - 8) ON/OFF and TRIPPED indicating position of operating handle;
 - 9) Operating handle to assume a centre position when tripped;
 - 10) Individual permanent thermal and magnetic trip elements in each pole;
 - 11) Electronic Trip unit for overload, overcurrent and earth fault with indication;
 - 12) Automatically opens all poles when overcurrent occurs on one pole;
 - 13) Test button on cover;
 - 14) Calibrated for 40 degrees C ambient, unless shown otherwise; and
 - 15) Do not provide single-pole circuit breakers with handle ties where multi-pole circuit breakers are shown.
 - (ii) Miniature Circuit Breakers (MCB):
 - 1) Moulded case, thermal-magnetic, quick make, and quick break, trip-free, of the indicating type showing OFF/ON and tripped position of the operating handle;
 - 2) In compliance with IEC 60947-1 and IEC 60947-2;
 - 3) Ultimate Breaking Capacity (Icu): 25 kA (minimum) at 415 V, a.c.;
 - 4) Ics not less than 75% of Icu; and
 - 5) Multi-pole circuit breakers designed to automatically open all poles when an overload occurs on one pole.
 - (iii) Residual Current Circuit Breaker (RCCB):
 - 1) Moulded case circuit breaker with electromagnetic release in case of earth fault;
 - 2) Trip Rating: 30 mA, 100 mA, 300 mA or 500 mA as required unless otherwise shown on drawings or DB Schedules;

- 3) Positive ON/OFF indication;
 - 4) TRIP indication on front face by red flag; and
 - 5) Tropicalized.
- (iv) Residual Current Circuit Breaker with Over Current Protection (RCBO):
- 1) The RCBO shall comply with IEC 61009-1 with any amendments made thereto;
 - 2) RCBO shall be voltage independent and shall not require any external supply;
 - 3) Moulded case circuit breaker with both thermal overcurrent protection and electromagnetic release in case of earth fault;
 - 4) Protection against overcurrent shall be in accordance with IEC 60898 with tripping characteristics of B, C or K according to the load applications;
 - 5) Trip Rating on Earth Fault: 30 mA, 100 mA, 300 mA, 500 mA as required unless otherwise shown on drawings or DB Schedules;
 - 6) The short circuit breaking capacity shall be minimum 25 kA or higher depending upon the short circuit level of the switchboard;
 - 7) RCBO shall be double pole type and shall have positive ON/OFF indication;
 - 8) TRIP indication on front face by red flag;
 - 9) The casing shall be of tamper-proof construction with casing designed from insulating material for rugged usage;
 - 10) Tropicalized;
 - 11) RCBO shall have all accessories for MCB and RCCB;
 - 12) Cable termination shall be from top and bottom with access from front and shall be suitable for use without any distinction to be made between supply side and load side;
 - 13) Contacts shall be silver alloy coated or equivalent;
 - 14) Live parts shall be protected against unintentional contact having IP 20 degree of protection;
 - 15) RCBO shall have resistance against nuisance tripping due to an impulse voltage or transient overvoltage; and
 - 16) The thermal tripping device shall be calibrated at a reference temperature of 30-degree C to IEC 60898. Appropriate correction factors shall be provided for other operating temperatures.
- (v) Enclosure: Suitable for location installed, IEC 60529, IP 31, minimum.
- (vi) Interlock: Enclosure and circuit breaker to prevent opening cover with circuit breaker in ON position.

- (g) Fused Switch Disconnecter, Individual, up to 600 Volts
 - (i) Conform to standards IEC 60947-1 and IEC 60947-3;
 - (ii) Rated insulation Voltage: 690 volts;
 - (iii) Rated Short-Circuit Current with Fuse Links: 80 kA;
 - (iv) Quick-make, quick-break, motor rated, load-break, heavy-duty (HD) type with external markings clearly indicating ON/OFF positions;
 - (v) Fuse mountings shall accept only current-limiting fuses;
 - (vi) Enclosure: Suitable for location installed, IEC 60529, IP 68, minimum; and
 - (vii) Interlock: Enclosure and switch to prevent opening cover with switch in ON position. Provide bypass feature for use by qualified personnel.
- (h) Non-Fused Switch Disconnecter, Individual, up to 600 Volts
 - (i) Conform to standards IEC 60947-1 and IEC 60947-3;
 - (ii) Quick-make, quick-break, motor rated, load-break, heavy-duty (HD) type with external markings clearly indicating ON/OFF positions. The disconnecter shall be able to withstand locked rotor current of the motor and able to open without any damage to the disconnecter;
 - (iii) Rated insulation Voltage: 690 volts;
 - (iv) Enclosure: Suitable for location installed, IEC 60529, IP 68, minimum;
 - (v) Interlock: Enclosure and switch to prevent opening cover with switch in the ON position. Provide bypass feature for use by qualified personnel; and
 - (vi) Equip with auxiliary switches (1 NO, 1 NC) rated 5 amperes at 230 volts for remote indication of position and interlocks. Additional auxiliary switches shall be provided as required by the application.
- (i) Fuse, up to 600 Volts
 - (i) Conform to standard IEC 60269;
 - (ii) Voltage Rating: As appropriate for application;
 - (iii) Rated Breaking Capacity: minimum 100 kA; and
 - (iv) Provide to fit mountings specified with switches.
- (j) Pushbutton, Indicating Light, and Selector Switches
 - (i) Conform to IEC 60947-5-1;
 - (ii) Voltage Rating: 250 volts;
 - (iii) Selector Switch Operating Lever: Standard. All Selector Switch shall be three position type;
 - (iv) Indicating Lights: 24 volts AC/DC CLUSTERED LED type, 1 watt maximum:
 - 1) ON: Green;
 - 2) OFF (NOT RUNNING): Red;
 - 3) POWER ON: White; and

- 4) TRIP: Amber.
- (v) Pushbutton Colour
 - 1) RUN, ON or START: Green;
 - 2) STOP, OFF or STOP: Red; and
 - 3) RESET: Black.
- (vi) Pushbuttons and selector switches lockable in OFF position where indicated;
- (vii) Legend Plate
 - 1) Material: Aluminium;
 - 2) Engraving: 11 character/spaces on one line, 14 character/spaces on each of two lines, as required, indicating specific function; and
 - 3) Letter Height: Manufacturer's standard.
- (k) Local Control Station (LCS)
 - (i) Large stainless-steel box, as previously specified, with hinged cover, except for those located in a hazardous location.
 - (ii) The complete assembly shall be rated IP65 (minimum) except for those that are specified to include a data jack.
 - (iii) Pushbuttons, indicating lights, isolators and selector switches shall be provided as shown on Drawing- P&ID Legend- Local Control Stations.
 - (iv) Where "Data Jacks" are shown to be provided in an LCS, they shall meet all applicable requirements of the general specifications.
 - (v) LCS installed in hazardous locations shall have explosion protection compatible with the zone classified.
 - (vi) Control station to include non-fused switch disconnecter where shown on the Drawings:
 - 1) Non-fused switch disconnecter to be lockable in the OFF position;
 - 2) Disconnect switches shall be provided for the motors rated ≤ 75 kW;
 - 3) Disconnect switch for VFD driven motors shall be provided with normally open early break contact to turn off VFD;
 - 4) Interlock control station door with switch disconnecter. Switch disconnecter to be OFF for door to be opened;
 - 5) Switch capable of disconnecting the motor when in a stalled condition; and
 - 6) Provide sufficient auxiliary contacts, in addition to those specified for remote indication of position, to completely isolate all pushbuttons, heaters, control circuits, etc.
 - (vii) Terminal Blocks
 - 1) Separate connection point for each power conductor 6 mm² or smaller and all control conductors entering or leaving box; and
 - 2) Spare Terminal Points: 25 %, eight, minimum.

- (viii) Wiring
 - 1) Type: 600-volts stranded copper;
 - 2) Size: For current to be carried, not less than 2.5 mm²; and
 - 3) Wiring and control devices operating at less than 50 volts AC or DC shall be isolated from 230- and 400-volts circuits by metal barriers.
- (ix) Wire Identification
 - 1) Numbered and tagged at each termination; and
 - 2) Wire Tags: As specified in the general specifications.
- (x) Nameplates and Legend Plates
 - 1) Permanently mount a nameplate on each local control station;
 - 2) Nameplate shall bear entire tag number found on the drawings;
 - 3) Nameplates to be engraved, rigid, laminated plastic type with adhesive back;
 - 4) Colour: White with black letters. Letter height 15 mm; and
 - 5) All control devices shall be identified by a legend plate as previously specified.
- (xi) Construction: Local control stations shall be manufactured and wired in an approved panel fabrication shop.
- (l) Terminal Junction Box
 - (i) Enclosure
 - 1) Large stainless-steel junction box with hinged cover unless otherwise indicated;
 - 2) Large cast metal junction box for IP 68 applications;
 - 3) Glass fibre reinforced polyester resin or 316 SST, both suitably rated for Ex de for-power applications in hazardous locations; and
 - 4) Glass fibre reinforced polyester resin or 316 SST, both rated Ex ia for intrinsically safe systems applications.
 - (ii) Terminal junction boxes rated IP 68 shall be cast metal; and
 - (iii) Terminal Blocks
 - 1) Separate connection point for each conductor entering or leaving box; and
 - 2) Spare Terminal Points: 25 %, minimum eight.
- (m) Terminal Block (up to 600 Volts)
 - (i) Conform to IEC 60947-7-1;
 - (ii) Rated Voltage: 600 volts, AC;
 - (iii) Size components to allow insertion of necessary wire sizes;

- (iv) Capable of termination of control circuits entering or leaving equipment, panels, or boxes;
- (v) Screw clamp compression, dead front barrier type, with current bar providing direct contact with wire between compression screw and yoke;
- (vi) Yoke, current bar and clamping screw of high strength and high conductivity metal;
- (vii) Yoke shall guide all strands of wire into terminal;
- (viii) Current bar shall ensure vibration-proof connection;
- (ix) Terminals
 - 1) Capable of wire connections without special preparation other than stripping;
 - 2) Capable of jumper installation with no loss of terminal or rail space; and
 - 3) Individual, rail mounted.
- (x) Marking system, allowing use of pre-printed or field-marked tags.
- (n) Magnetic Control Relay
 - (i) Conform to standard IEC 60947-1;
 - (ii) Voltage Rating: 690 volts;
 - (iii) Rated Operating Current at 230 volts: 3-amperes, minimum;
 - (iv) Latching Attachment: Mechanical latch having unlatching coil and coil clearing contacts; and
 - (v) All control relays shall have indicating lights when energized.
- (o) Time Delay Relay
 - (i) Conform to standard IEC 61812-1 or IEC 61439;
 - (ii) Solid-state electronic, field convertible ON/OFF delay;
 - (iii) One normally open and one normally closed contact (minimum) rated 5 amperes at 250 volts;
 - (iv) Repeat accuracy plus or minus 2 %; and
 - (v) Timer adjustment from 1 to 60 seconds, unless otherwise indicated on the Drawings.
- (p) Elapsed Time Meter
 - (i) Drive: Synchronous motor;
 - (ii) Range: 0 to 99,999.9 hours, non-reset type;
 - (iii) Mounting: Semi-flush, panel; and
 - (iv) Contacts: 5 amps, 230 volts.
- (q) Magnetic Contactor
 - (i) Conform to standard IEC 60947-1;
 - (ii) Electrically operated, electrically held;

- (iii) Main Contacts
 - 1) Voltage Rating: 690 volts.
 - 2) Power driven in one direction with gravity dropout.
 - 3) Silver alloy with wiping action and arc quenchers.
 - 4) Continuous-duty, rated as required.
 - 5) Two-, three- and four-pole as shown on the Drawings and / or functionality required.
- (iv) Control: As shown on the Drawings or as required;
- (v) One normally open and one normally closed auxiliary contact rated 10 amperes at 230 volts; and
- (vi) Degree of Protection: As required for location of installation, IP 31, minimum.
- (r) Phase Monitor Relay
 - (i) Voltage and phase monitor relay shall drop out in 6 seconds, or less on low voltage, voltage unbalance, loss of phase or phase reversal;
 - (ii) Provide a magnetic control relay with additional contacts where required for application;
 - (iii) Adjustable trip voltage and time delay settings;
 - (iv) Transient Protection: 10,000 volts;
 - (v) Enclosure: IP 31, minimum; and
 - (vi) Mounting: Multi-pin plug-in socket base.
- (s) Magnetic Lighting Contactor
 - (i) Comply with IEC 60947;
 - (ii) Electrically operated by dual-acting, single coil mechanism; and
 - (iii) Inherently interlocked, electrically held in, and mechanically held in both OPEN and CLOSED position.
- (iv) Main Contacts
 - 1) Double-break, continuous-duty, rated 20 / 30 / 60 amperes as per application, 600 volts, withstand rating of 22,000 amps rms symmetrical at 250 Volts or 14,000 amps rms symmetrical at 480 Volts as per application;
 - 2) Marked for electric discharge lamps, tungsten, and general-purpose loads;
 - 3) Position not dependent on gravity, hooks, latches, or semi-permanent magnets;
 - 4) Capable of operating in any position; and
 - 5) Visual indication for each contact.

- (v) Auxiliary contact relay for two-wire Form 3 control.
- (vi) One normally open and one normally closed auxiliary contact rated 10 amperes continuous;
- (vii) Fully rated neutral terminal;
- (viii) Provision for remote pilot lamp with use of auxiliary contacts; and
- (ix) Clamp type, self-rising terminal plates for solderless connections.
- (t) Enclosures: See Article Large Stainless Steel Box
- (u) Industrial Capacitors
 - (i) Comply with IEC;
 - (ii) Enclosed, indoor, dustproof, three-phase capacitor units containing internally mounted, indicating type, high interrupting-capacity, current-limiting fuses and discharge resistors;
 - (iii) Units containing PCB dielectric fluid are unacceptable; and
 - (iv) kVAr Ratings
 - 1) kVAr ratings of capacitors connected to individual motor circuits were selected based on expected motor power factor;
 - 2) Check motor nameplate and manufacturer's power factor and no-load current data for actual motor installed; and
 - 3) Reduce capacitor kVAr if required, so the size does not exceed the motor manufacturer's recommended maximum size, and so it does not exceed the value required to raise motor no-load power factor to 0.95.
- (v) Dry Type, Shielded, Isolation Transformer
 - (i) Enclosure: IEC 60529, IP 31, minimum;
 - (ii) Primary Winding Taps: Four, 2-1/2 %, full capacity; two above and 2 below rated voltage;
 - (iii) Insulation System: Manufacturer's standard;
 - (iv) Core and coils to not exceed rated temperature rise at rated ambient with neutral current 200 % of phase current; and
 - (v) Isolation and Noise Suppression
 - 1) Electrostatic Shield (Including Core and Coils): One foil type, located between primary and secondary windings;
 - 2) Earth Lead: From each shield to unit enclosure;
 - 3) Common Mode Attenuation: 120 dB;
 - 4) Transverse Mode Attenuation: 20 dB;
 - 5) Impedance: 4.3 %, maximum;
 - 6) Reactance: Minimum 3 %; maximum 4.2 %;
 - 7) Regulation, No-Load to Full-Load: Plus or minus 1.2 %; and
 - 8) Output Distortion: None added.

- (w) Low Voltage, Secondary Surge Protective Equipment
 - (i) Conform to IEC 60099;
 - (ii) Surge Capacitor
 - 1) Impregnated with non-PCB, biodegradable dielectric fluid; and
 - 2) Integral discharge resistor which will drain residual voltage to 50-volts crest in less than 1 minute after disconnection from circuit.
 - (iii) Arrestor: High strength metal oxide valve elements enclosed in high strength, corrosion-resistant, moulded resin housing;
 - (iv) Equip capacitor and arrestor with mounting nipple, flat washer, and nut suitable for knockout or bracket mounting;
 - (v) Surge Protective Device (SPD) where specified shall be in accordance with IEC 61643-11, installed and coordination provided in accordance with IEC 61643:
 - 1) SPD Type 1 where specified shall be provided on all 4 conductors (i.e. 3 Phases and Neutral) of rating 25 kA;
 - 2) SPD Type 2 where specified shall be coordinated with upstream Type 1 SPD and provided on all 4 conductors (i.e. 3 Phases and Neutral) of minimum rating 40 kA;
 - 3) ALL SPD shall have End of Life indicator, Safety Reserve system, online pluggable type and with volt-free contacts for remote indication;
 - 4) Each SPD operation signal shall be sent to SCADA through Energy Meters or separate I/Os. Necessary I/Os and cabling shall be provided regardless of whether they are shown on the drawings or interface schematics;
 - 5) Each SPD shall be protected by suitably rated MCB; and
 - 6) Performance
 - The surge protection device shall provide protection in both Common Mode (Phase to Ground & Neutral to Ground) and differential Mode (Phase to Neutral); and
 - The surge protection device shall be designed to perform and survive in lightning intense environment and frequent switching transient. It must also provide full protection against overvoltage for all possible transient modes e.g. Phase to Neutral, Phase to Earth, Neutral to Earth.
 - The surge protection device shall have continuous indication of its protection status. The status indication should clearly show Full protection present, reduced protection – replacement required, and No protection – failure of surge protection device.
- (x) Medium Voltage Motor Surge Protection
 - (i) Conform to IEC 60099;

- (ii) Surge Capacitor
 - 1) Synthetic film and paper, dielectric impregnated with non-PCB biodegradable dielectric fluid, internal discharge resistor and three-pole, hermetically-sealed bushings, rated 25 kV; and
 - 2) Integral discharge resistor which will drain residual voltage to 50-volts crest in less than 5 minutes after disconnection from circuit.
- (iii) Surge Arrestors: Station class, valve type, metal oxide, and single-pole.
- (y) Support and Framing Channels
 - (i) Carbon Steel Framing Channel
 - 1) Material: Rolled, mild strip steel, 12-gauge/2.66 mm, in compliance with BS EN 10088; and
 - 2) Finish: Electro-galvanised after fabrication.
 - (ii) Paint-Coated Framing Channel: Carbon steel framing channel with electro-deposited rust inhibiting acrylic or epoxy paint;
 - (iii) PVC-Coated Framing Channel: Carbon steel framing channel with 1 mm polyvinyl chloride coating;
 - (iv) Stainless Steel Framing Channel: Rolled, Type 316 L stainless steel, 12-gauge/2.66 mm;
 - (v) Extruded Aluminium Framing Channel
 - 1) Material: Extruded from Type 6063-T6 aluminium alloy; and
 - 2) Fittings fabricated from Alloy 5052-H32.
 - (vi) Non-Metallic Framing Channel
 - 1) Material: Fire retardant, fibre reinforced vinyl ester resin;
 - 2) Channel fitting of same material as channel, and
 - 3) Nuts and bolts of long glass fibre reinforced polyurethane.
- (z) Intrinsic Safety Barrier
 - (i) General: Intrinsic safety barrier shall provide a safe energy level for exposed wiring in a hazardous area when circuit is connected to power source in non-hazardous area;
 - (ii) Rating: Power source shall be rated 24 volts DC, nominal, with not more than 250 volts available under fault conditions;
 - (iii) Contact Rating: 5 amperes, 250 volts AC; and
 - (iv) Mounting: Rail or surface.
- (aa) Nameplates
 - (i) Material: Laminated plastic (Trefolite);
 - (ii) Attachment Screws: Stainless steel;
 - (iii) Colour: White, engraved to a black core;

- (iv) Engraving
 - 1) Pushbuttons/Selector Switches: Name of drive controlled on one, two, or three lines, as required; and
 - 2) Panel boards: Panel board designation, service voltage, and phases.
- (v) Letter Height
 - 1) Pushbuttons/Selector Switches: 3 mm; and
 - 2) Panel boards: 10 mm.
- (bb) Switchboard Mat
 - (i) General: Supply, deliver and install insulated mat manufactured in accordance with IEC 61111 having a breakdown of 40 kV minimum (class 4) for 22 kV system and 20 kV minimum (class 2) for 6.6 kV and 415 V systems.
- (cc) Cable/Conduit Sealing Device
 - (i) General: Sealing device for use where needed to seal conduit or cable penetrations of walls, floors, or ceilings;
 - (ii) Prevent the passage of smoke, toxic fumes, and water before and after exposure to flame;
 - (iii) Material: Fire-resistant, water-proof elastomeric material;
 - (iv) Thickness: 100 mm, minimum; and
 - (v) Cable penetration sealing material shall comply with local codes and FSSD requirement, work shall be carried out by authorized applicator. Necessary registration and liaison with authorities is included in the scope of works under this contract.

G5.10.3 Execution

- (a) General
 - (i) Install equipment in accordance with manufacturer's recommendations;
 - (ii) Use appropriate conduit and conductor entry fittings with enclosures to maintain the specified enclosure environmental capability after installation;
 - (iii) Label all equipment to identify power source feeding it;
 - (iv) All incoming and outgoing cables shall be labelled with cable tags; and
 - (v) Labelling to be approved by S.O..
- (b) Outlet and Device Boxes
 - (i) Install suitable for conditions encountered at each outlet or device in wiring or raceway system. Installation shall meet applicable requirements of CP 5.
 - (ii) Size
 - 1) Depth: Minimum 35 mm unless otherwise required by structural conditions. Box extensions not permitted:
 - Hollow Masonry Construction: Install with sufficient depth such that conduit knockouts or hubs are in masonry void space.

- 2) Ceiling Roses: Minimum 50 mm diameter sheet steel device box, unless otherwise required for installed fixture;
 - 3) Switch and Receptacle: Minimum 86 x 86 mm (for single outlet) or 86 x 146 mm (for twin outlet) sheet steel device box for concealed applications; and
 - 4) Install cast metal boxes of similar size for all exposed applications.
- (iii) Locations
- 1) To avoid interference with mechanical equipment or structural features, relocate outlets as directed by S.O..
 - 2) Light Switch: Install on lock side of doors; and
 - 3) Light Fixture: Install in symmetrical pattern according to room layout.
- (iv) Mounting Height
- 1) General:
 - Dimensions given to centreline of box; and
 - Where specified heights do not suit building construction or finish, mount as directed by S.O..
 - 2) Light Switch: 1,200 mm above floor;
 - 3) Telephone Outlet: 150 mm above counter tops or 350 mm above floor;
 - 4) Wall-Mounted Telephone Outlet: 1,400 mm above floor;
 - 5) Power Socket Outlet
 - General Interior Areas: 350 mm above floor;
 - General Interior Areas (Counter Tops): Install device plate bottom or side flush with top of splashback or 150 mm above counter tops without splashback;
 - Industrial Areas, Workshops: 1,000 mm above floor; and
 - Outdoor, All Areas: 500 mm above finished grade.
 - 6) Special-Purpose Receptacle: 500 mm above floor or as shown on the Drawings.
- (v) Install plumb and level.
- (vi) Flush Mounted
- 1) Install with concealed conduit in finished areas;
 - 2) Install proper type extension rings or plaster covers to make edges of boxes flush with finished surface; and
 - 3) Holes in surrounding surface shall be no larger than required to receive box.
- (vii) Support boxes independently of conduit by attachment to building structure or structural member.

- (viii) Threaded studs driven in by powder charge and provided with lock washers and nuts are acceptable in lieu of expansion shields.
- (ix) Provide plaster rings where necessary.
- (x) Boxes embedded in concrete or masonry need not be additionally supported.
- (xi) Install galvanised mounting hardware in industrial areas.
- (xii) Install separate junction boxes for flush or recessed lighting fixtures where required by fixture terminal temperature.
- (xiii) Boxes Supporting Fixtures: Provide means of attachment with adequate strength to support fixture.
- (xiv) Open no more knockouts in sheet steel device boxes than are required; seal unused openings.
- (xv) Box Type (Steel Raceway System)
 - 1) Interior Dry Locations:
 - Exposed Rigid Conduit: Cast metal;
 - Concealed Raceways: Sheet steel;
 - Concrete Encased Raceways: Cast metal; and
 - Lighting Circuits, Ceiling: Sheet steel.
 - 2) Interior Wet Locations
 - Exposed Raceways: Cast metal;
 - Concealed Raceways: Cast metal up to 1,000 mm above floor, sheet steel elsewhere;
 - Concrete Encased Raceways: Cast metal; and
 - Lighting Circuits, Ceiling: Sheet steel.
 - 3) Cast-In-Place Concrete Slabs: Sheet steel.
- (xvi) Box Type (Rigid Aluminium Raceway System): Cast aluminium.
- (xvii) Box Type (Non-Metallic Raceway System)
 - 1) Corrosive Locations: Non-metallic;
 - 2) Exposed Raceways: Non-metallic;
 - 3) Concealed Raceways: Non-metallic; and
 - 4) Concrete Encased Raceways: Non-metallic.
- (c) Junction Boxes and Pull Boxes
 - (i) Install where shown and where necessary to terminate, tap-off or redirect multiple cable or conduit runs;
 - (ii) Install pull boxes where necessary in raceway system to facilitate conductor installation;
 - (iii) Install in conduit runs at least every 40 metres or after the equivalent of two right-angle bends;

- (iv) Use outlet boxes as junction and pull boxes wherever possible and allowed by applicable codes;
- (v) Installed boxes shall be accessible;
- (vi) Do not install on finished surfaces;
- (vii) Install plumb and level;
- (viii) Support boxes independently of cable or conduit by attachment to building structure or structural member;
- (ix) Install bar hangers in frame construction or fasten boxes directly as follows:
 - 1) Wood: Wood screws;
 - 2) Concrete or Brick: Bolts and expansion shields;
 - 3) Hollow Masonry Units: Toggle bolts; and
 - 4) Steelwork: Machine screws.
- (x) Threaded studs driven in by powder charge and provided with lock washers and nuts are acceptable in lieu of expansion shields;
- (xi) Boxes embedded in concrete or masonry need not be additionally supported;
- (xii) At or Below Grade
 - 1) Install boxes for below grade conduit flush with finished grade in locations outside of paved areas, roadways or walkways;
 - 2) If adjacent structure is available, box may be mounted on structure surface just above finished grade in accessible but unobtrusive location;
 - 3) Obtain S.O.'s written acceptance prior to installation in paved areas, roadways or walkways; and
 - 4) Use boxes and covers suitable to support anticipated weights.
- (xiii) Flush Mounted
 - 1) Install with concealed conduit;
 - 2) Holes in surrounding surface shall be no larger than required to receive box; and
 - 3) Make edges of boxes flush with final surface.
- (xiv) Mounting Hardware
 - 1) Noncorrosive Dry Areas: Galvanised;
 - 2) Noncorrosive Wet Areas: Stainless steel; and
 - 3) Corrosive Areas: Stainless steel.
- (xv) Location/Type
 - 1) Finished, Indoor, Dry: IEC 60529, IP 21;
 - 2) Unfinished, Indoor, Dry: IEC 60529, IP 31;

- 3) Unfinished, Indoor and Outdoor, Wet: IEC 60529, IP 66
 - Steel Raceway System: Cast metal;
 - Rigid Aluminium Raceway System: Cast aluminium; and
 - Cable System: Cast aluminium unless otherwise indicated.
 - 4) Indoor and Outdoor, Wet and Corrosive: IEC 60529, IP 66
 - PVC-Coated Rigid Electro-Galvanised Steel Raceway System: Stainless steel;
 - Non-Metallic Raceway System: Non-metallic; and
 - Cable System: Non-metallic.
 - 5) Indoor and Outdoor, Hazardous: IEC 60079-10, suitable for Zone 1 and Zone 2 hazardous locations
 - Steel Raceway System: Cast metal;
 - Rigid Aluminium Raceway System: Cast aluminium; and
 - Cable System: Cast aluminium unless otherwise indicated.
 - 6) Outdoor Locations: IEC 60529, IP 66, Non-metallic;
 - 7) Submerged Locations: IEC 60529, IP 68; and
 - 8) Industrial Use in Areas Not Otherwise Classified: IEC 60529, IP 55, unless otherwise indicated on the drawing.
- (xvi) Install drain/breather fittings in IEC 60529, IP 65 and 66 enclosures.
- (d) Telephone Terminal Cabinet
- (i) Install with top of cabinet 2 metres above floor; and
 - (ii) Minimum Installed Door Opening: 120 degrees.
- (e) Telephone Outlet
- (i) Supply, install, test and commission outlet boxes with an RJ45 receptacle and cover plates meeting requirements of IMDA.
- (f) Wiring Devices
- (i) Switches
 - 1) Mounting Height: See Article Outlet and Device Boxes;
 - 2) Install with switch operation in vertical position; and
 - 3) All switches shall be mounted so knob is in the raised position when the switch is OFF and depressed when the switch is ON.
 - (ii) Power Socket Outlets
 - 1) Install with earth slot down;

- 2) Weatherproof Socket Outlet:
 - Weatherproof sockets shall be switched;
 - Install in cast metal box; and
 - Provide screw-on cap with captive chain.
 - 3) Special-Purpose Receptacles: Install in accordance with manufacturer's instructions.
- (g) Device Plates
- (i) Securely fasten to wiring device; ensure a tight fit to box;
 - (ii) Flush Mounted: Install with all four edges in continuous contact with finished wall surfaces without use of mats or similar materials. Plaster fillings will not be acceptable;
 - (iii) Surface Mounted: Plate shall not extend beyond sides of box, unless plates have no sharp corners or edges; and
 - (iv) Install with alignment tolerance to box of 2 mm.
- (h) Local Control Stations
- (i) Install for all motors as specified regardless of whether they are shown on the drawings;
 - (ii) Wall or pedestal mounts as specified or to suite site condition as directed by S.O.;
 - (iii) Terminate all cable entries in the bottoms of the enclosures; and
 - (iv) Cable entries to be sealed with cable gland fittings.
- (i) Terminal Junction Box
- (i) Install in accordance with Article Junction and Pull Boxes;
 - (ii) Label each block and terminal with permanently attached, non-destructible tag;
 - (iii) Do not install on finished outdoor surfaces; and
 - (iv) Location/Type: As specified for Junction and Pull Boxes.
- (j) Dry Type Transformer (0- to 600-Volts Primary)
- (i) Load external vibration isolator such that no direct transformer unit metal is in direct contact with mounting surface;
 - (ii) Provide moisture-proof, flexible conduit for electrical connections;
 - (iii) Connect voltage taps to achieve (approximately) rated output voltage under normal Plant load conditions;
 - (iv) Provide wall brackets for single-phase units, 15 to 167-1/2 kVA and three-phase units, 15 to 112 kVA; and
 - (v) Isolation Transformer: Earth isolation shields to unit enclosure with conductor of same material, and same size minimum, as shield ground lead provided with unit.

- (k) Support and Framing Channel
 - (i) Install where required for mounting and supporting electrical equipment and raceway systems;
 - (ii) Channel Type
 - 1) Interior, Wet or Dry (Noncorrosive) Locations
 - Aluminium Raceway: Extruded aluminium;
 - PVC-Coated Conduit: PVC coated; and
 - Steel Raceway and Other Systems Not Covered: Carbon steel or paint coated.
 - 2) Interior, Corrosive (Wet or Dry) Locations
 - Aluminium Raceway: Extruded aluminium;
 - PVC Conduit: BS EN 10088, Type 316L stainless steel or non-metallic; and
 - PVC-Coated Steel Conduit and Other Systems Not Covered: BS EN 10088, Type 316L stainless steel, non-metallic, or PVC-coated steel.
 - 3) Outdoor, Noncorrosive Locations
 - Steel Raceway: Carbon steel or paint-coated framing channel, except where mounted on aluminium handrail, then use aluminium framing channel; and
 - Aluminium Raceway and Other Systems Not Covered: Aluminium framing channel.
 - 4) Outdoor Corrosive Locations
 - Conduit: BS EN 10088, Type 316L stainless steel or non-metallic;
 - Aluminium Raceway: Aluminium; and
 - PVC-Coated Steel Conduit, and Other Systems Not Covered: BS EN 10088, Type 316L stainless steel, non-metallic, or PVC-coated steel.
 - (iii) Paint cut ends prior to installation with the following:
 - 1) Carbon Steel Channel: Zinc-rich primer;
 - 2) Painted Channel: Rust-inhibiting epoxy or acrylic paint;
 - 3) Non-Metallic Channel: Epoxy resin sealer; and
 - 4) PVC-Coated Channel: PVC patch.
- (l) Intrinsic Safety Barriers
 - (i) Install in compliance with SS, IEC, or BS;
 - (ii) Arrange conductors such that wiring from hazardous areas cannot short to wiring from non-hazardous area; and
 - (iii) Stencil “INTRINSICALLY SAFE CIRCUIT” on all boxes enclosing barriers.

- (m) Switchboard Matting
 - (i) Install 1,000 mm width in front of all switchgear, switchboard, motor control centres, and panel boards; and
 - (ii) Matting shall run full length of all sides of equipment that have operator controls or afford access to devices.
- (n) Cable/Conduit Sealing Device
 - (i) Install in strict accordance with manufacturer's instructions;
 - (ii) Sealing devices shall effectively seal all openings in concrete walls, floor, and ceilings; and
 - (iii) Cables and conduits shall be held tightly by sealing devices.
- (o) Motor Surge Protection
 - (i) Earthing in accordance with CP 5, SS 551 and IEC or BS;
 - (ii) Low Voltage: Earth terminals to equipment bus; and
 - (iii) Medium Voltage
 - 1) Install surge arrestor in enclosure adjacent to motor junction box; and
 - 2) Earth terminals to motor frame and motor circuit earthing conductor.
- (p) Environmental Protection and Selection of Material
 - (i) Interior Finished / Unfinished Dry locations: Epoxy coated electro-galvanized steel or Type 304 stainless steel;
 - (ii) Interior Wet locations: Type SS 316L stainless steel; and
 - (iii) Outdoor locations: Type SS 316L stainless steel.

END OF SECTION